

Math 407 Section 1
Fall 2025
Final Exam
May 14, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- If the problem asks for a proof, be sure to carefully justify your work, including any theorems from class.

Note: you may NOT use a theorem or result from class to prove something when it makes the problem entirely trivial. If you are unsure whether a particular theorem or result is allowed, just ask!

Unless otherwise stated, all rings will be commutative with identity and all ring homomorphisms will send the identity to the identity.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

(a) The discriminant of a cubic polynomial can tell us if its roots are all real

(b) If $F \subseteq K$ is a field extension, then $\text{Gal}_F(K) = [K : F]$

(c) $\mathbb{Q}(\sqrt[3]{5})$ is a splitting field

(d) The ring $\mathbb{Z}[\sqrt{-5}]$ is a UFD

(e) Any subgroup of a cyclic group is also cyclic

2. (10 points)

(a) State the definition of a principal domain

(b) State the third isomorphism theorem for groups

(c) Write down two non-isomorphic non-Abelian groups of order 8

3. (10 points)

(a) Give an example of a field extension of $\mathbb{Q}$ of degree 2

(b) Give an example of a field extension of $\mathbb{Q}$ which is transcendental

(c) Give an example of a field extension of $\mathbb{Q}$ which is not normal

(d) Give an example of a field extension of $\mathbb{Q}$ whose Galois group is isomorphic to S_3

4. (10 points)

Consider the group

$$G = \mathbb{Z}_{12} \times \mathbb{Z}_{12}.$$

(a) Determine all the elements in the subgroup H generated by $(1, 1)$ and $(0, 3)$.

(b) Explicitly describe the cosets of G/H . How many are there?

(c) What standard group is G/H isomorphic to?

5. (10 points)

(a) State the Structure Theorem for Finite Abelian Groups

(b) Write the group $\mathbb{Z}_{35} \times \mathbb{Z}_{15} \times \mathbb{Z}_{100}$ in invariant factor form

(c) how many Abelian groups of order 2^6 are there, up to isomorphism?

6. (10 points)

Consider the set

$$R = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \text{ odd} \right\} \subseteq \mathbb{Q}.$$

(a) Prove that the set R is a subring of $\mathbb{Q}$.

(b) Prove that $\frac{a}{b} \in R$ is a unit if and only if a is odd.

(c) Prove that the principal ideal $(2) \subseteq R$ is a maximal ideal.

7. (10 points)

Let $\mathbb{Q} \subseteq K \subseteq \mathbb{C}$ be a splitting field of $(x^2 - 2)(x^2 - 3)$.

(a) Prove that $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $[K : \mathbb{Q}] = 4$

(b) Explain why the Galois group $\text{Gal}_{\mathbb{Q}}(K)$ has four elements

(c) Prove that if $\sigma \in \text{Gal}_{\mathbb{Q}}(K)$ then $\sigma(\sqrt{2}) = \pm\sqrt{2}$ and $\sigma(\sqrt{3}) = \pm\sqrt{3}$.

(d) Prove that $\text{Gal}_{\mathbb{Q}}(K)$ is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$